

The Cerebellum Optimizes Perceptual Predictions about External Sensory Events

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Summary

Perception and action are governed not only by sensory information but also by prior predictions about sensory events. These sensory predictions allow one to react more rapidly to predictable information in the environment [1] and to perceptually distinguish self-produced and externally produced sensations [2–6]. In order to be accurate, however, all sensory predictions need continuous recalibration to match the changing properties of the environment, the sensorimotor system, or both. Earlier studies showed that the cerebellum is crucial for the recalibration of sensory predictions capturing the sensory consequences of one's motor behavior [5, 7]. Here we asked whether the cerebellum, a structure intimately linked to plasticity within the motor domain [8–13], also accounts for the recalibration of sensory predictions about external sensory events within the perceptual domain in a nonmotor task. Cerebellar patients and healthy controls were equally able to predict the time of reappearance of a moving target that temporarily disappeared behind an occluder. However, patients were significantly impaired in recalibrating this spatiotemporal prediction to account for an experimentally added delay. This suggests that the cerebellum plays a domain-general role in fine tuning predictive models.

Results

Sensory information alone is not sufficient for producing reliable perception and behavior. Instead, the brain also forms “sensory predictions” based on prior information and generalized knowledge about the likely outcome given a certain situation [14–17]. For example, when emptying a ketchup bottle by hitting it with one's hand, the other hand, which holds the bottle, generates sufficient grip force to compensate the self-generated hitting load without delay. Mechanisms driven purely by sensory feedback are too slow to allow such motor compensation, which hence must rely on predictive processes building on efferent motor commands [16, 17]. Similarly, to perceptually estimate whether a car might cross one's way and to react in time, the sensorimotor system would benefit from predicting the car's upcoming movement based on afferent sensory input [18]. In order to be accurate, however, sensory predictions need constant recalibration to match the

ever-changing properties of one's sensorimotor system and environment [3, 13]. Such recalibration is needed irrespective of whether sensory predictions are related to motor behavior, perception, or both. But which brain structure is responsible for recalibrating sensory predictions?

In agreement with the role of the cerebellum in processing and adapting motor-related information [8–12, 19–21], we and others have demonstrated that the cerebellum is needed to optimize self-action and self-action perception by recalibrating predictions capturing the sensory consequences of one's actions [5, 7, 9]. Given the high share of connections between the cerebellum and cerebrocortical areas not directly related to motor behavior, and given the accumulating number of reports implying cerebellar contributions beyond the motor domain [22, 23], we hypothesized that the cerebellum might additionally be needed for recalibrating predictions exclusively related to the perceptual domain, i.e., even in the absence of self-action. If true, this would support a domain-general role of the cerebellum in recalibrating predictive models.

To test our hypothesis, we conducted a “baseline” and a “recalibration” experiment estimating, respectively, subjects' ability to use sensory predictions perceptually and to keep these predictions precisely tuned. The study included 15 cerebellar patients (see [Table S1](#) available online) and 15 age-matched controls with no neurological or psychiatric disorders ([Supplemental Experimental Procedures](#)). All subjects gave written informed consent according to the Declaration of Helsinki prior to the experiment, and the study was approved by the local ethics committee. Subjects' performance depended on the spatiotemporal prediction of a sensory event, namely, the time of reappearance of a moving target. During each trial, a visual target moved from left to right on a monitor and disappeared in the middle of its trajectory behind an occluder ([Figure 1A](#)). Although the speed of the visual target was always constant ($\sim 16^\circ/\text{s}$), the time it needed to pass behind the occluder was manipulated across trials. Subjects were instructed to constantly fixate a central fixation cross in the middle of the occluder and to decide whether the reappearance of the target at the right side of the occluder was “too early” or “too late,” assuming that it maintained constant speed ([Figure 1A](#)). Using this procedure, we aimed at probing a spatiotemporal sensory prediction exclusively within the perceptual domain.

In the baseline experiment, we assessed subjects' ability to accurately predict the time of target reappearance. We investigated subjects' perceptual reports while varying the difference between the actual and the theoretical reappearance time (200 ms; [Figure 1B](#)). This difference Δt was equally distributed over ± 180 ms. [Figure 2A](#) depicts the course of a typical baseline experiment as well as the related fitted psychometric function, plotting the probability of “too late” responses as a function of Δt . The reliability of each subject's sensory predictions was estimated using two measures derived from the corresponding psychometric functions. First, we estimated the point of subjective equivalence (PSE) between predicted and actual Δt . At the PSE, subjects perceive the target reappearance as “right on time” and thus respond

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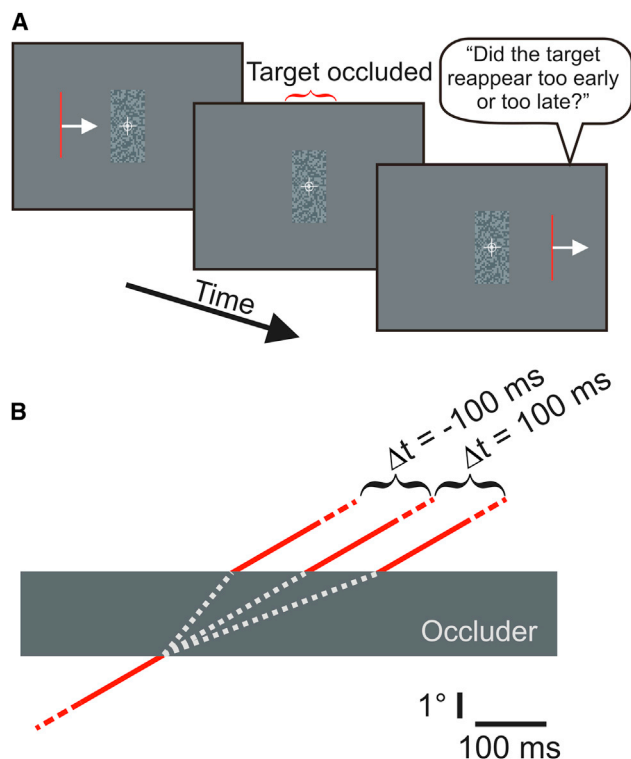


Figure 1. Experimental Task Design

(A) A visual target (red bar) moved from the left side of the stimulus screen to the right. In the middle of its movement trajectory (middle panel), it disappeared under an opaque occluder. The speed of the target ($\sim 16^\circ/\text{s}$) was always constant when the target was visible. However, the time the target needed to pass the occluder was varied experimentally. After the target had reappeared at the right end of the occluder, subjects had to manually indicate whether the target had reappeared “too early” or “too late,” given its constant speed.

(B) The time of target reappearance Δt was varied relative to the duration the target would need to pass the occluder when moving at constant speed (i.e., 200 ms, middle target trajectory). Negative Δt (left exemplary trajectory) denotes that the target reappeared earlier as compared to a continuous movement with constant speed, whereas positive Δt (right trajectory) denotes that the target reappeared later.

50% “too early” and 50% “too late” across trials. Second, we assessed the just noticeable difference (JND) for changes in Δt . The JND was defined as the difference in Δt between the PSE and a hit rate of 75%. Importantly, the quality of the underlying psychometric fit to the data (quantified by residuals) did not differ between groups [baseline: $t(24) = 1.37$, $p = 0.18$; recalibration: $t(27) = 0.94$, $p = 0.36$].

In the baseline experiment, the mean Δt at the PSE (bPSE) was 57 ± 10 ms in cerebellar patients and 30 ± 9 ms in healthy controls (Figure 3A), indicating that both groups slightly overestimated the time the target needed to pass the occluder. Across groups, the bPSE was not significantly different [$t(28) = -1.90$, $p = 0.07$; t test]. The JND was also not significantly different between groups ($z = -0.19$, $p = 0.85$; Wilcoxon rank-sum test). It amounted to a median value of 55 ms ($Q1 = 49$ ms, $Q3 = 79$ ms) in patients and 55 ms ($Q1 = 51$ ms, $Q3 = 64$ ms) in controls (Figure 3B).

In the recalibration experiment, the target reappeared 100 ms after individual’s baseline PSE in half of the trials (“recalibration trials”). The remainder of the randomly interleaved trials was used to estimate the recalibrated PSE

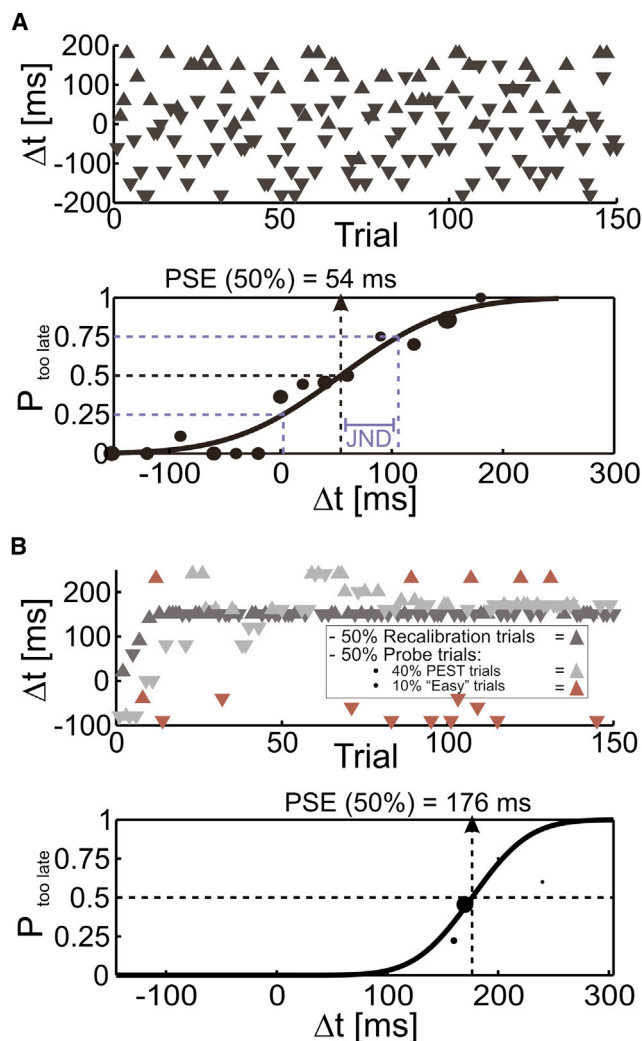


Figure 2. Results of an Exemplary Control Subject

(A) In the baseline experiment, the levels of Δt were equally distributed across trials, ranging between ± 180 ms, as shown in the upper panel. Triangles pointing upward represent “too late” responses; triangles pointing downward represent “too early” responses. The lower panel exhibits the corresponding probability of “too late” responses for a given Δt . The fit of the psychometric function indicates a baseline point of subjective equivalence (bPSE at $P_{\text{too late}} = 0.5$) of $\Delta t = 54$ ms and a just noticeable difference (JND) of $\Delta t = 52$ ms.

(B) In the recalibration experiment, the target reappeared 100 ms later than the bPSE (i.e., $\Delta t = 154$ ms for this subject) in 50% of trials, namely in the so-called recalibration trials (dark gray triangles). The remaining 50% of randomly interleaved probe trials were used to calculate the actual point of subjective equivalence (recalibration point of subjective equivalence; rPSE). Eighty percent of these probe trials were PEST trials (light gray triangles) in which Δt was governed by an adaptive staircase procedure (PEST) that approximated the PSE. Twenty percent of probe trials were “easy” trials (brown triangles) with Δt s above the perceptual threshold. Recalibration trials induced a rightward shift of the corresponding psychometric function, resulting in an rPSE of 176 ms (lower panel).

For further information on the different trial types and the calculation of the psychometric function, see [Supplemental Experimental Procedures](#). For all individual subjects’ PSEs, see [Figure S1](#).

(rPSE; Figure 2B). If recalibration trials induced an updating of the time of predicted target reappearance, this should lead to a rightward shift of the psychometric function as

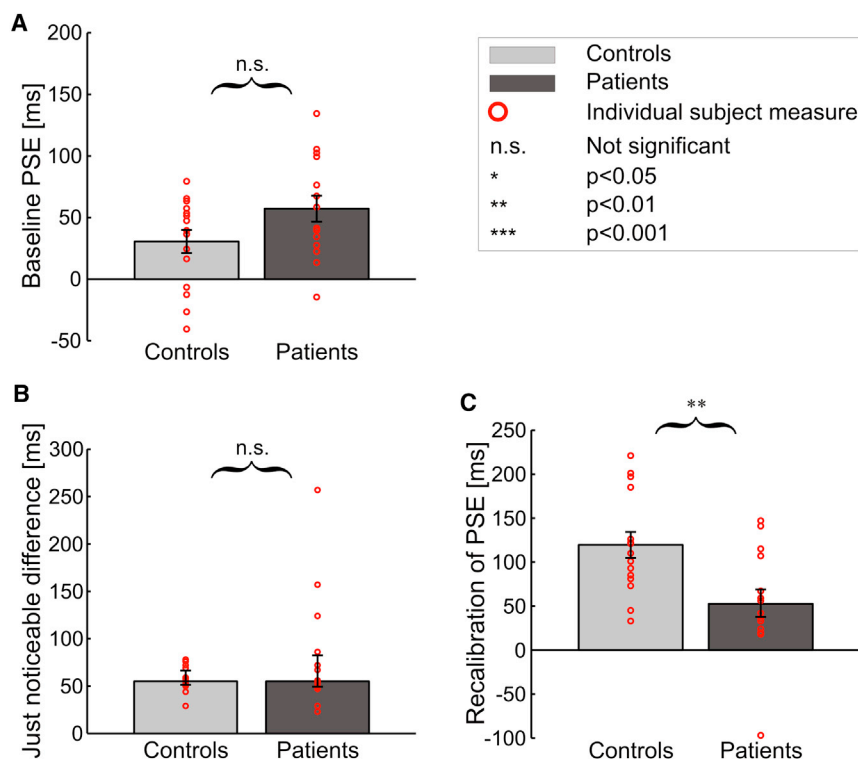


Figure 3. Group Results of Patients and Controls

(A) The baseline point of subjective equivalence (bPSE) was not significantly different between groups, although patients exhibited a slightly larger bPSE (mean: controls 30 ms, patients 57 ms). Error bars indicate $\pm$ SEM.

(B) The just noticeable difference (JND), as a measure of perceptual accuracy, did not show a significant difference between both groups (median 55 ms in both groups). This indicates that both groups were equally able to form accurate sensory predictions. Error bars indicate the inter-quartile range.

(C) Recalibration, i.e., the difference between rPSE and bPSE, showed a highly significant difference ($p = 0.002$) between groups, indicating an impairment of cerebellar patients in recalibrating their sensory predictions (mean: controls 120 ms, patients 52 ms). Error bars indicate $\pm$ SEM.

For group results after exclusion of possible outliers, see Figure S2.

compared to baseline, and thus to a higher rPSE. This shift is illustrated in Figure 2 for a control subject. The corresponding increase in rPSE as compared to the bPSE suggests recalibration of a sensory prediction in this subject (also see Figure S1).

As a group, controls changed their sensory prediction in the recalibration experiment: compared to an average bPSE of 30 ± 9 ms, the mean of the rPSEs increased to 150 ± 19 ms. Hence, recalibration amounted to 120 ± 15 ms and fully accounted for the 100 ms delay [from which it was statistically indistinguishable; $t(14) = 1.32$, $p = 0.208$; one-sample t test]. In patients, however, recalibration was significantly smaller than in controls [$t(28) = 3.11$, $p = 0.002$; t test] and amounted to only 53 ± 16 ms (bPSE = 57 ± 10 ms, rPSE = 110 ± 21 ms; Figure 3C). This implies that patients' recalibration of a sensory prediction was significantly impaired and reduced by 56% relative to controls. Importantly, this group difference in recalibration was neither carried by nor limited to patients with degenerative cerebellar diseases ($n = 12$). It was also apparent in the remaining three subjects with acute unilateral focal lesions: each showed a significantly smaller recalibration compared to the control group [#1: $t(14) = 3.55$, $p = 0.003$; #3: $t(14) = 5.21$, $p < 0.001$; #6: $t(14) = 6.69$, $p < 0.001$; one-sample t test, Bonferroni corrected]. Finally, we did not find any significant correlation between the behavioral measures acquired in the baseline experiment and recalibration (bPSE $\times$ rPSE-bPSE, JND $\times$ rPSE-bPSE) in either group ($p \geq 0.05$).

To rule out that differences in recalibration could be explained by eye movements, we analyzed (1) eye position, (2) velocity, and (3) number of saccades within a 400 ms time window starting 100 ms before target disappearance (Figure 4). For each of these measures, we performed a 2×2 ANOVA with the between-subject-factor subject group and the within-subject-factor experiment. In a separate set of analyses, we additionally included the within-subject-factor time bin, with the 400 ms epoch divided into four 100 ms

bins ($2 \times 2 \times 4$ ANOVA) to also account for transient changes in eye movements. None of these ANOVAs returned a significant difference between groups or interaction effects ($p \geq 0.05$; Figure 4). Furthermore, linear regression analyses

between each eye movement parameter and the amount of recalibration revealed no significant correlations ($p \geq 0.05$; Figure S3A).

Discussion

We showed that cerebellar patients and controls were equally accurate in predicting the time of reappearance of a moving target that temporarily disappeared behind an occluder. However, patients were significantly impaired in recalibrating this spatiotemporal prediction to account for a temporal delay added in the recalibration experiment. We suggest that the cerebellum is generally needed to keep sensory predictions precisely tuned—even within the perceptual domain.

Previous research has highlighted the role of the cerebellum in motor control and learning [8, 10]. This involvement in the motor domain has often been discussed within the context of internal forward models (FMs) [8, 9, 11, 12, 20, 21, 24–27]. FMs capture the expected sensory consequences of one's actions based on efference copies [4, 6] of motor commands and enable fast and accurate movements despite delayed or missing sensory feedback [17, 28]. Accordingly, cerebellar lesions lead to less predictive control of motor behavior [11, 25–27], a finding that could also explain various clinical symptoms [8]. Moreover, its characteristic crystalline circuitry implicates the cerebellum in establishing and updating FMs. The cytoarchitecture of a cerebellar microcomplex allows it to adjust the transfer of information from its mossy fiber inputs to its output structures in response to a teaching signal provided by climbing fibers in a temporally precise fashion [29–31]. A microcomplex could thereby learn to mimic the input-output characteristic of any neural system with which it shares common input [9, 28]. Such circuitry seems ideal to, for instance, form a predictive model of the motor plant and to recalibrate this FM to account for the plant's changing

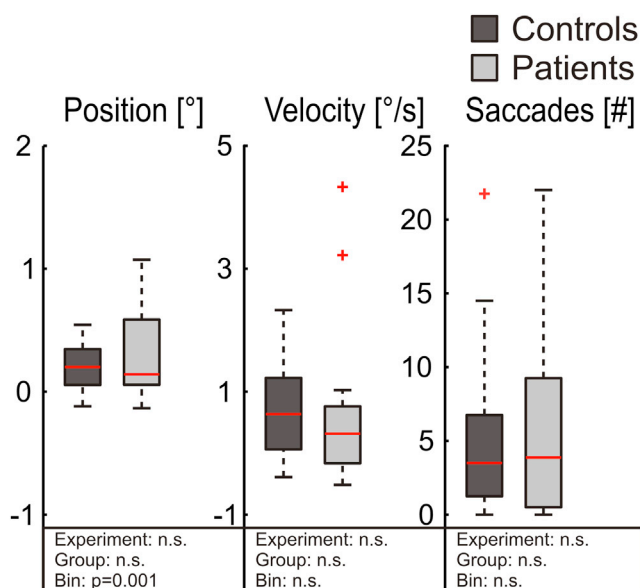


Figure 4. Comparison of Eye Movements between Patients and Controls
To rule out that the observed difference in recalibration was due to eye movements, we tested for group differences in horizontal eye position, eye velocity, and the number of fixational saccades, respectively. Specifically, we ran ANOVAs with the factors group and experiment (and time bin). There were no significant group effects or interactions ($p \geq 0.05$; the respective significance of each main factor is depicted in each panel; n.s. = not significant). This is further supported by the box plots for the three eye movement parameters shown here. There are no differences between patients (light gray) and controls (dark gray). The central red bars reflect the median of each measure, the vertical extent of each box captures the range between the 25th (Q1) and the 75th (Q3) percentile, and the whiskers extend to extreme values [except putative outliers “+,” i.e., values $> Q3 + 1.5 (Q3 - Q1)$ or $< Q1 - 1.5 (Q3 - Q1)$]. Note that putative outliers were not excluded from statistical comparisons. See also [Figure S3](#).

properties (due to fatigue, aging, or disease), with the efferent motor command as common input, the predicted sensory action consequences as an output, and the prediction error as a teaching signal [28]. Other studies further highlight the role of the cerebellum in the acquisition and updating of FMs relevant for the perceptual distinction of self-produced versus externally produced sensory events [5, 9, 24, 32]. Finally, the cerebellum also enables learning to predict an upcoming sensory event based on other (predictive) sensory information in order to trigger appropriate responses and reflexes in a temporally precise fashion [31, 33–35].

All of the aforementioned evidence can be generally understood as a cerebellar contribution to the learning of predictive models related to motor behavior—preliminarily supporting the view of the cerebellum as a motor structure [19]. But although such learning is clearly needed in the motor domain to guarantee accurate performance, it is also required to precisely tune predictions to the statistics of the “sensory environment”—even if predictions are solely restricted to the perceptual domain. Given the cerebellum’s circuit properties and its interconnections also with cerebral areas engaged in higher-order perceptual processing and spatiotemporal awareness (e.g., posterior parietal cortex) [22, 23], we hypothesized that the cerebellum could guarantee the updating of sensory predictions across domains, i.e., also in nonmotor tasks. Our patients’ deficit in recalibrating a spatiotemporal prediction for perception clearly supports this notion. Such a

domain-general learning mechanism could likewise explain other nonmotor impairments of cerebellar patients, e.g., in processing predictive sequences [36, 37] or in forming “cognitive associations” [38].

Given the characteristic properties of cerebellar processing [9, 24, 30, 31, 36, 39], our results could be more specifically explained by patients’ inability to adjust the timing of a sensory prediction to account for the experimentally induced prediction error in the temporal domain. Alternative explanations, such as a recalibration of perceived target speed or space (occluder width), appear to be less likely also because these parameters were experimentally fixed and thus perhaps more resistant to adaptive changes due to their higher relative reliability [40]. Ultimately, however, our experiment cannot distinguish the detailed level (or levels) at which recalibration occurred. However, and more important for our interpretation, our task design aimed to restrict recalibration to the sensory domain, in that the perceptual decision in our task did not depend on any motor component.

One might argue that a motor component was still needed to signal subjects’ perceptual decision. However, this response was made only after the target had reappeared and therefore could not interfere with the formation of the sensory prediction. In addition, we tried to exclude any indirect motor effects resulting from eye movements by instructing central fixation. Moreover, we made sure that any residual eye movements did not differ between patients and controls. We also could not detect any correlation between eye movement parameters and the amount of recalibration irrespective of a subject group. This is not to say, however, that there are no oculomotor deficits in cerebellar patients. Rather, the oculomotor demands of our experiment—namely, central fixation—prevented the oculomotor deficits of our patients from surfacing [41]. Finally, the fact that both groups performed equally well in the baseline experiment further rules out impairments in basic processes such as timing, motion processing, or attention that potentially could have likewise explained the observed recalibration deficit in cerebellar patients ([Supplemental Discussion](#)).

Comparable baseline performance rather raises the question of whether the cerebellum is needed at all for storing predictive models of the type required in our task. Seemingly, in all of our patients (with the possible exception of one or two putative outliers; [Figure S2](#)) the cerebellum was needed only when predictions required updating. While several patient studies probing sensory predictions in the perceptual domain have reported similar results [5, 7, 42], many other studies in the motor domain also report general FM deficits in baseline performance [25–27]. This discrepancy could be explained by differing task requirements. For instance, it could be that, in the earlier set of studies and in our own experiment, intact baseline performance could simply be explained by an additional representation of a predictive model in cerebral cortex that supports these more cognitive tasks [28]. In fact, several studies imply a representation of predictive (mental) models in posterior parietal cortex [43–45], and it has been suggested that these models—as opposed to those in the cerebellum—support aspects of sensory prediction more closely related to consciousness [43]. The cerebellum’s particular role in such a context could instead be the unconscious fine tuning of sensory predictions.

The posterior lobe of the cerebellum seems a likely candidate for the recalibration of sensory predictions relevant for the perceptual domain [24, 32]. This agrees with our study, in that the three patients with focal cerebellar lesions all

suffered from acute damage to the posterior cerebellar lobe, the commonly affected region being lobule VIII in the left cerebellar hemisphere.

In conclusion, we propose a domain-general role of the cerebellum in fine tuning predictive models.

Supplemental Information

Supplemental Information includes three figures, one table, Supplemental Discussion, and Supplemental Experimental Procedures and can be found with this article online at <http://dx.doi.org/10.1016/j.cub.2013.04.027>.

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Supplemental Information

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Perceptual Predictions

about External Sensory Events

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Supplemental Inventory

Figure S1. Individual Points of Subjective Equivalence. *Related to Figure 2.*

Figure S2. Group Results after Exclusion of Possible Outliers. *Related to Figure 3.*

Figure S3. Eye Movements and Recalibration. *Related to Figure 4.*

Table S1. Etiology of Disease, Lesion Description, and Ataxia Severity

Supplemental Discussion

Supplemental Experimental Procedures

Supplemental References

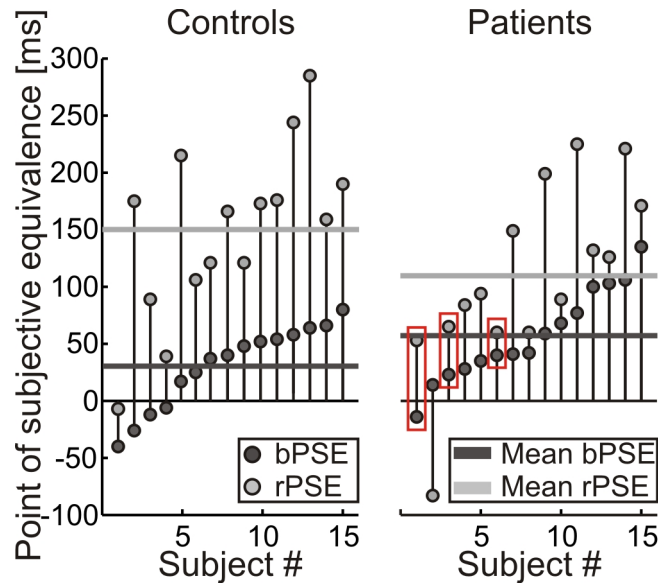


Figure S1. Individual Points of Subjective Equivalence during Baseline and Recalibration Experiment, Related to Figure 2

Individual and mean baseline points of subjective equivalence (bPSE; dark gray) and recalibration points of subjective equivalence (rPSE; light gray) of controls are depicted in the left panel and those of patients in the right panel, respectively. Red rectangles indicate patients that suffered from acute focal lesion. The bPSEs of both groups were not significantly different ($p \geq 0.05$, t-test) although patients showed on average a slightly higher bPSE (mean bPSE patients: 57 ms; mean bPSE controls: 30 ms). However, there was a strong significant difference in the amount of recalibration, i.e. the difference between bPSE and rPSE, between both groups ($p = 0.002$, t-test). Relative to the baseline PSE controls showed a much larger PSE in the recalibration experiment than did the patient group.

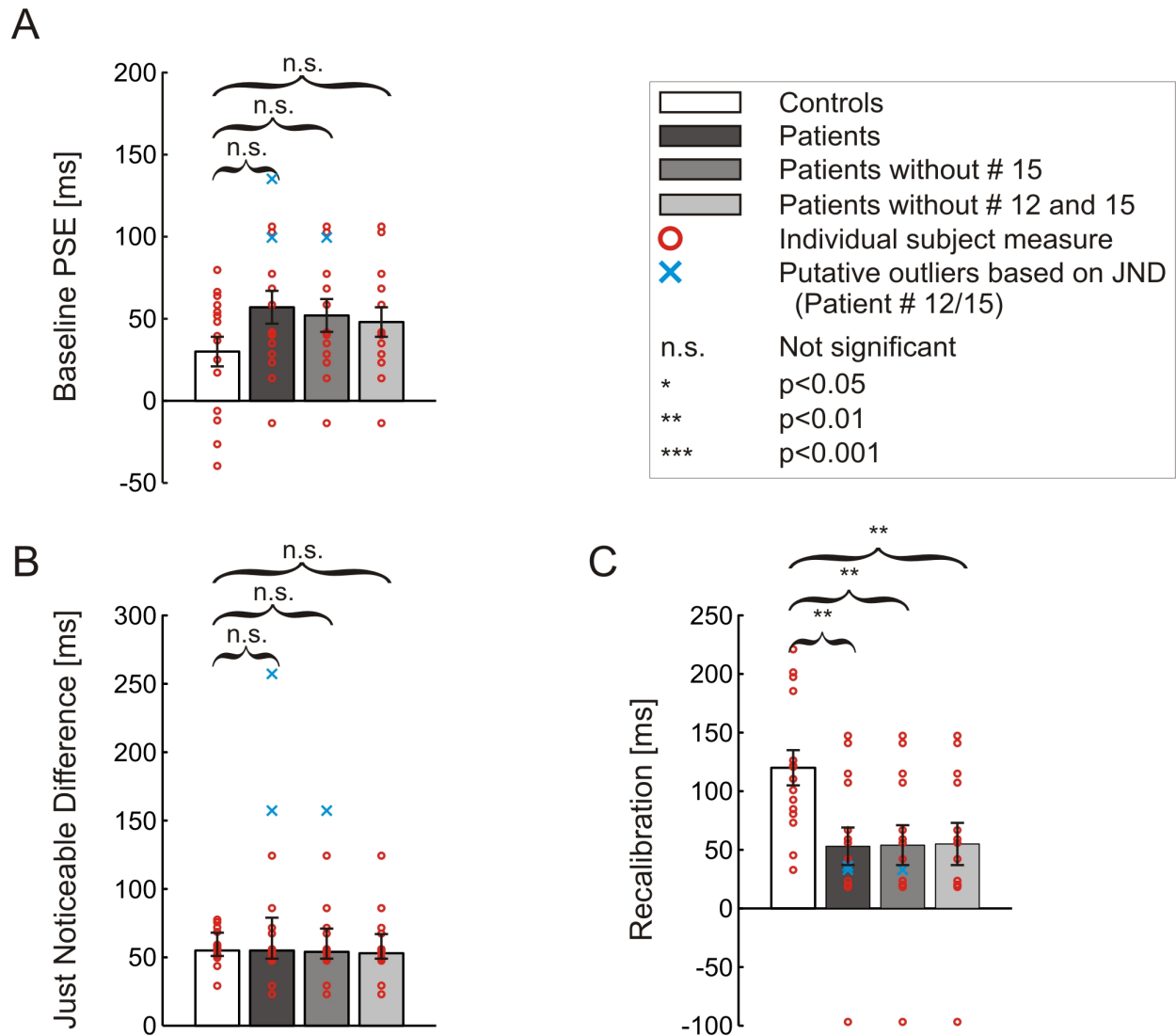


Figure S2. Group Results after Exclusion of Possible Outliers Based on Their JND, Related to Figure 3

Average group results before and after exclusion of two possible outliers based on the JND (patients #12 and #15, blue crosses).

(A and B) The mean baseline point of subjective equivalence (bPSE $\pm$ SEM) (A) and the median just noticeable difference (JND $\pm$ interquartile range) (B) were not significantly different between groups also when either one (JND $\leq Q1 - k[Q3 - Q1]$ or $\geq Q3 + k[Q3 - Q1]$; $k=3$) or two ($k=1.5$) patient(s) with the largest JND(s) was/were excluded. In fact, the bPSE and the JND of the patient group even became more similar to that of controls numerically if these putative outliers were removed.

(C) Recalibration was always significantly different between patients and controls no matter whether or not we excluded the putative outliers ($p < 0.01$). Error bars indicate means $\pm$ SEM.

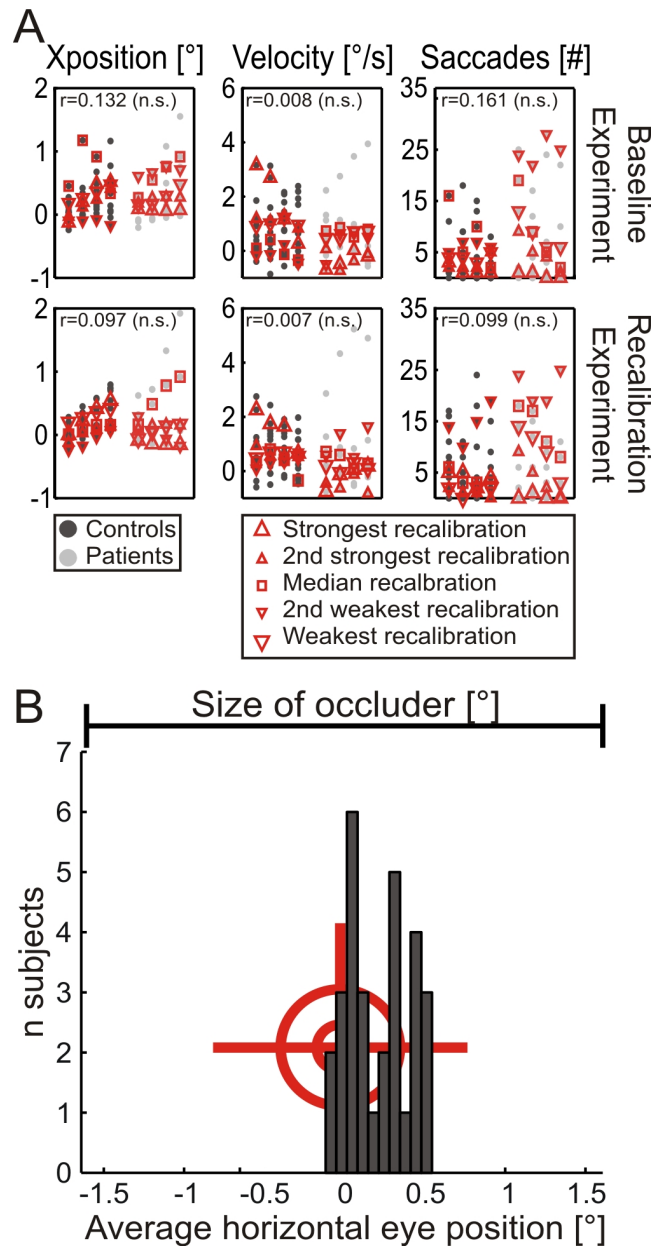


Figure S3. Eye Movements and Recalibration, Related to Figure 4

(A) Horizontal eye position, eye velocity, and saccade number for the subjects with the strongest, the second strongest, the median, the second weakest, and the weakest recalibration are shown. Individual columns refer to the four 100ms time bins considered (see Results section for details). There were no apparent relationships between eye movements and the strength of recalibration. There likewise were no significant correlations between these three eye movement parameters and the amount of recalibration, neither in the baseline nor in the recalibration experiment ($p \geq 0.05$, corresponding r -values are provided in individual panels, n.s. = not significant).

(B) This figure depicts a histogram of all subjects' average horizontal eye position relative to the fixation cross. The x-axis extends over the width of the occluder (3.2°). Average fixation was always well within the limits of the occluder.

Table S1. Etiology of Disease, Lesion Description, Ataxia Severity, and Recalibration

Demographics			Lesion Description			SARA (Scale for the Assessment and Rating of Ataxia)										Behavioral Measures	
Patient #	Age	Sex	Disease Duration*	Global (g), Focal (f)	Diagnosis	G (8)	S (6)	Si (4)	SD (6)	FC (4)	N (4)	F (4)	H (4)	Sum (40)	JND [ms]	Recalibration [ms]	
1	26	m	5 days	f	left paramedian cerebellar hemorrhage	2	2	0	0	0.5	0	0.5	0.5	5.5	49	67	
2	49	m	8	g	ISCA	2	3	0	2	1.5	2	2	2	14.5	86	-97	
3	34	f	4 days	f	ischemia PICA left	1	1	0	0	0.5	0.5	0	1	4	53	42	
4	69	f	7	g	ISCA	2	1	0	0	1	0	0.5	0	4.5	23	56	
5	62	f	3	g	ISCA	2	2	0	1	1	1	1	1	9	53	59	
6	67	f	3 days	f	ischemia PICA left	3	1	0	0	0.5	0.5	1	0	6	67	20	
7	69	m	13	g	ADCA	5	3	0	1	1	0.5	2	2	14.5	29	107	
8	33	m	13	g	ISCA	2	2	0	2	1	0	1	0	8	55	18	
9	74	f	2	g	ISCA	2	1	0	3	1	1.5	1.5	1	11	56	141	
10	53	f	4	g	ISCA	2	1	0	0	1	1	1	2	8	124	20	
11	75	f	5	g	ISCA	7	4	0	3	1	1.5	1.5	2	20	47	147	
12	42	f	2	g	ISCA	2	3	0	1	1	0	2.5	1.5	11	157	33	
13	50	m	20	g	ISCA	2	1	1	2	1	0	2	2	11	72	24	
14	45	m	2	g	ISCA	2	1	1	1	1	0	1.5	1	8.5	50	115	
15	63	f	6	g	SCA6	0	0	0	0	0	0	1	1	2	257	36	

ISCA denotes idiopathic sporadic cerebellar ataxia; PICA denotes posterior inferior cerebellar artery; ADCA denotes autosomal-dominant cerebellar atrophy; SCA denotes spinocerebellar atrophy.

M denotes male; f denotes female.

G, gait; S, stance; Si, sitting; SD, speech disturbance; FC, finger chase; N, nose-finger test; F, fast alternating hand movements; H, heel-shin slide. JND, just noticeable difference.

The score that would indicate maximal impairment on each subtest is given in parentheses. We did not find any significant correlation between the sum of the SARA scores and any of the behavioral measures ($p \geq 0.05$).

* Disease duration in years, unless specified otherwise.

Supplemental Discussion

How do our results relate to non-motor deficits that have been described in cerebellar patients such as motion processing or timing? It has been suggested that patients with cerebellar disease lack accurate timing mechanisms in learning and performing actions [1, 2], and in time perception itself [3]. Cerebellar contributions to timing of perceptual events were also reported in an fMRI Study by O'Reilly and co-workers [4], in which subjects had to make spatiotemporal predictions in a task that was similar to our experiment. In their study, the cerebellum was the only brain region that was activated when making spatiotemporal predictions as compared to forming purely spatial predictions. Therefore, the authors suggest that the cerebellum predicts the timing of perceptual events. Yet, although our task required sensory predictions that represent a precise temporal relationship between events [5], our results failed to provide evidence for any general timing deficit since patients were equally able to accurately predict the time of target reappearance in our baseline experiment. How can this seeming discrepancy to the aforementioned findings be reconciled? A possible explanation for the different results is that here the timing information was embedded in a visual motion stimulus while the earlier patient reports describing impaired time perception used auditory stimuli with longer temporal delays. In the case of the fMRI study we suggest that fMRI activity could also reflect the spatiotemporal error between a predicted and an actual sensory state, a notion that is also supported by the results of other groups [6, 7].

Another deficit of cerebellar patients that has been consistently reported is an impairment in visual motion perception [8-12]. In case of an impaired perception of the moving target in our task, subjects should also be unable to accurately predict its reappearance even in the baseline condition. This was clearly not the case. Yet, earlier studies used global visual motion stimuli and perceptual judgments that markedly differed from our task (i.e. the perceptual tracking of a single target). Given these differences in experimental protocols and the resulting differences in the underlying neuronal processing, it thus may not be surprising that we do not find deficits that could simply be explained by impaired motion processing per se. We rather suggest that patients' deficit in our task refers to their inability to optimize a predictive model about an externally moving target. In fact, Cerminara et al. [13] provided direct electrophysiological evidence for an internal model of an externally moving target in the cat cerebellum. Moreover, these authors discuss that the origin or a replica of the observed neural correlate of this sensory-based prediction in the cerebellum could very well be represented elsewhere in the brain. Therefore, the specific contribution of the cerebellum could be to refine sensory predictions if necessary. The latter considerations are elaborated in more detail in the discussion of the main manuscript.

Supplemental Experimental Procedures

Subjects

The patient group was comprised of fifteen patients (six men, nine women; average 54 years; standard deviation ± 16 years). Twelve patients suffered from global cerebellar degeneration ($n = 1$ autosomal-dominant cerebellar atrophy [ADCA], $n = 1$ spinocerebellar atrophy type 6 [SCA6], $n = 10$ idiopathic sporadic cerebellar ataxia [ISCA]). All patients were examined by a specialist on degenerative ataxias (M.S.). None of them showed any clinical, electrophysiological or imaging evidence of extra-cerebellar disease involvement. Secondary causes of ataxia (e.g. ataxias of inflammatory, vascular or metabolic origin) were excluded in all patients by serum and CSF analysis and by MRI imaging. Out of the patients with genetically confirmed ataxia, only patients with SCA6 were included since degeneration in this SCA subtype is known to be essentially limited to the cerebellum [14]. In the ADCA and ISCA patients, SCA1,2,3,7 and 17 were excluded, as these SCA subtypes are known to affect several noncerebellar CNS structures. The three remaining patients had unilateral focal lesions due to posterior inferior cerebellar artery (PICA) stroke ($n = 2$) or paramedian hemorrhage ($n = 1$), all located in the left cerebellar hemisphere. All of these patients with vascular lesions were studied 3 to 5 days after stroke. Ataxia severity was assessed using the Scale for the Assessment and Rating of Ataxia (SARA [15]). Refer to supplemental table S1 for more detailed patient information.

The control group consisted of 15 age-matched healthy controls (six men, nine women; average 51 years; standard deviation ± 13 years) with no known neurological or psychiatric disorders. Accordingly, patients and controls did not differ in age ($p \geq 0.05$, t-test).

All subjects had normal or corrected-to-normal visual acuity and gave written informed consent according to the declaration of Helsinki prior to the experiment. The study was approved by the local ethics committee.

Experimental Procedures

All experiments were realized using MATLAB R2007b (The MathWorks, Inc.) and the Cogent Graphics toolbox 1.29 (<http://www.vislab.ucl.ac.uk/cogent.php>). Stimuli were presented on a CRT screen (viewing distance 57 cm, frame rate 100 Hz, resolution 1280 x 1024 pixels, screen width 40.5 cm) and subjects' head was being stabilized by a head and chin rest. All experiments were performed in a darkened room to enhance stimulus visibility. During the experiment, participants' eye movements were registered at 50Hz using a video-based dark-pupil eye tracking technique (ViewPoint Eye Tracker; Arrington Research).

In all experiments, an occluder (width: 3.2° visual angle; height: 6.3°) was presented in the center of the otherwise gray stimulus screen (Fig. 1A). The occluder was composed of a random pattern of dark-gray and light-grey squares, which kept the overall luminance of the occluder identical to that of the grey background and thereby allowed minimizing the visibility of afterimages. The random arrangement of squares was updated at the beginning of every trial. A white fixation cross was depicted on top of the occluder and subjects were instructed to fixate this cross all the time. Each experimental trial started with a fixation interval of random duration, ranging between 0.5 to 1.5 s in steps of 100 ms. After that interval a red bar (width: 0.16° ; height: 5.7°) started to move horizontally from left to right with a constant velocity of $15.8^\circ/\text{s}$. The bar started to move from one out of four randomly chosen start positions (9.5° , 12.7° , 15.8° or 19° to the left of the screen center). During midflight the target disappeared behind the central occluder. After some time t the target reappeared from behind the occluder and continued

moving at the same constant velocity of $15.8^\circ/\text{s}$ to the right until it reached its end position (9.5° , 12.7° , 15.8° or 19° to the right of the screen center, respectively). Importantly, we manipulated the time t that the target spent behind the occluder: given its constant speed, the target could reappear at the right end of the occluder either at the correct time (i.e. after $t = 200$ ms), or too early (i.e. $t < 200$ ms), or too late (i.e. $t > 200$ ms). The amount of manipulation is referred to as the relative time of target reappearance Δt (Fig. 1B), namely the difference between the actual time of reappearance t and the correct time (200 ms). After the target had reappeared subjects had to judge within 10 s and in a two-alternative forced-choice fashion (2AFC) if the target had reappeared "too early" or "too late", respectively. Subjects reported their decision by pressing either the right ("too early") or the left ("too late") button of a response box placed in front of them. A trial was successfully completed when a subject provided the perceptual decision in time.

In the baseline experiment, we assessed subjects' perceptual decisions while Δt was varied according to the method of constant stimuli (MCS; $\Delta t = \pm 180, \pm 150, \pm 120, \pm 90, \pm 60, \pm 40, \pm 20$, or 0 ms). The actual amount of Δt was randomly selected by the MCS procedure in every single trial (Fig. 2A). We collected an overall of 150 trials. For each subject we calculated a psychometric function, capturing the probability P of responding "too late" for a given Δt , using a probit fit [16] (Fig. 2A). The point of subjective equivalence (PSE) between the expected and the actual time Δt of target reappearance was derived from the fitted probit function. This "baseline PSE" (bPSE) was defined by the amount of Δt at which a participant was responding at chance (50% "too early" and 50% "too late" responses). Furthermore, we estimated the just noticeable difference (JND) for our experimental manipulation by calculating the difference in Δt at the bPSE and at the point at which subjects responded "too early"/"too late" in 75% of the cases.

The recalibration experiment was identical to the baseline experiment except that the statistics of Δt were altered: While different levels of Δt were equally distributed in the baseline experiment and amounted to 0 ms on average (i.e. there was no temporal bias), Δt was biased towards large positive values in the recalibration experiment. In other words, in the majority of trials the target would reappear too late. Specifically, 50% of the trials in this experiment served as so-called recalibration trials (Fig. 2B). In these trials, Δt was set to an individual's bPSE + 100 ms, i.e. the target appeared 100 ms later than expected by this subject, given her/his baseline PSE. To reduce the possibility that subjects become aware of this temporal bias, the level of Δt in the recalibration trials was gradually increased over the first 10 consecutive trials before reaching its final level ($\Delta t_1 = \text{bPSE}$; $\Delta t_2 = \text{bPSE} + 10$ ms; $\Delta t_3 = \text{bPSE} + 20$ ms; etc.). The remainder of the randomly interleaved trials consisted of "probe trials" which were used to calculate subjects' psychometric function. These probe trial comprised of so-called "PEST trials" (40% of all trials) and "easy trials" (10% of all trials). In PEST trials, Δt -values were selected based on an adaptive staircase procedure, namely the "parameter estimation by sequential testing" (PEST) procedure [17]. For detailed information about the algorithm of this staircase procedure please refer to reference [17]. The specific PEST parameters that we chose for our recalibration experiment were: Δt at start level = - 80 ms; initial step size = - 160 ms; target step size = 0 ms (to prevent early termination of the procedure); target probability [$P_{\text{too late}}$] = 0.5; WALD-constant = 0.5; minimum Δt = - 180 ms; maximum Δt = 500 ms). Hence, this staircase procedure targeted subjects' recalibration PSE (rPSE) at which they responded at chance (i.e. $P_{\text{too late}} = 0.5$; Fig. 2B).

In easy trials, Δt -values were chosen in a way that should guarantee that these values were above subjects' perceptual thresholds (i.e. above their JND). They served to motivate subjects by reducing the amount of trials close to chance level and to provide additional data

points in the more peripheral range of subjects' psychometric function. The Δt of these trials was randomly set to either bPSE + 180 ms or otherwise - 40 ms, - 60 ms, or - 90 ms (Fig. 2B). During the recalibration experiment 150 trials were collected in total. For fitting the psychometric function we only used probe trials (PEST trials and easy trials) while excluding the first 30 trials of an experiment to allow prior recalibration. Recalibration was assessed by calculating the relative change of each subject's rPSE as compared to the baseline bPSE. In figure S1, the individual and mean PSEs for patients and controls from both experiments are shown.

Statistical Analysis

All perceptual measures (bPSE, JND, rPSE-bPSE) were first tested for an underlying normal distribution within each group using Lilliefors test. Except for the JND, all measures were normally distributed. In case of a normal distribution the mean $\pm$ the standard error of the mean (SEM) is reported for each respective measure and we performed two-sample t-tests (assuming unequal variances) for group comparisons. In the case of JNDs, which were not normally distributed, the median as well as the interquartile range between the first (Q1) and the third quartile (Q3) are reported, while we used the Wilcoxon rank-sum test for group comparisons.

Because inspection of the JND suggested possible outliers in the patient group, we further investigated this possibility and we also tested if outlier exclusion would change the qualitative outcome of our group comparisons. The number of detected outliers ranged from zero [18] over one ($\text{JND} \leq Q1 - k[Q3 - Q1]$ or $\geq Q3 + k[Q3 - Q1]$; $k=3$) to two ($\text{JND} \leq Q1 - k[Q3 - Q1]$ or $\geq Q3 + k[Q3 - Q1]$; $k=1.5$), depending on the statistical criterion applied. Importantly, the general pattern of results did not change if the patients identified as putative outliers were excluded from the overall analyses. If at all, their exclusion made the overall baseline performance even more similar to that of the control group in numerical terms while the difference in recalibration persisted (Figure S2). Therefore, we did not exclude any patient from any of our analyses reported in the main manuscript.

Finally, to further explore possible relationships between the acquired behavioral measures across experiments we correlated the bPSE and the JND of the baseline experiment with the amount of recalibration (rPSE-bPSE). This is especially important to test if, for example, the slightly, but non-significantly elevated baseline PSE in the patient group could contribute to the group difference in recalibration. Yet, we did not find any significant correlation in neither group ($p \geq 0.05$; uncorrected for multiple comparisons).

In addition to our perceptual measures, we controlled for group differences in horizontal eye position, horizontal eye velocity, and the number of saccades. All measures were calculated offline based on a low-pass filtered eye position record (2nd-order 10 Hz Chebyshev digital low-pass filter Type II, $R = 3$). Saccades were detected using a velocity threshold criterion ($v > 15^\circ/\text{s}$). We tested for group differences separately in each of the three eye movement parameters using analyses of variance (ANOVA; compare Results section for further details). While this set of analyses was performed on the overall number of trials, we repeated these analyses only for those trials in which the target reappeared at most 90 ms too early. With this we tried to more specifically probe for putative differences in predictive eye movements during the 200ms epoch the constantly moving target would need to pass the occluder but not for reactive eye movements that might have been caused by the reappearing bar (given an expected saccade latency of about 100ms or longer). The same qualitative results were obtained in this second set of analyses. Finally, we tested whether the eye movement parameters were correlated with the amount of recalibration using regression analyses (also compare Figure S3A). Figure S3B additionally shows the average horizontal eye position of all subjects from a 400 ms period starting 100 ms

before target disappearance, plotted relative to the occluder. This figure demonstrates that, on average, all subjects held fixation within the borders of the occluder. Due to a temporary unavailability of our eye tracking device the eye data of four patients (patients number 8,9,11, and 12) could not be recorded. However, there were no significant differences between the acquired perceptual measures between the subgroups of patients with and without eye movement data ($p \geq 0.05$; two-sample t-test assuming unequal variances).

Analyses were performed using MATLAB R2007b (The MathWorks, Inc.), SPSS 20 (IBM), and Microsoft Excel 2003.

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